# Activity: 1

## 1 Medical Image Enhancement:

A doctor receives an x-ray image where the bones are visible, but the darker tissue have poor visibility.

1. Which gray-level transformation would you recommend?

→ Log Transformation.

2. Why is it suitable?

→ Log transformation enhances low-intensity pixel values while compressing high-intensity values. This makes the darker tissues more visible

3. What effect will it have on the image?

Dark tissues become brighter and more visible.
Hidden details in low-intensity regions are enhanced.

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## 2. Night-Time Surveillance:

A CCTV image is very dark, with most pixel values between 0 and 40.

1. Which transformation would improve the visibility?

→ Log transformation.

2. Why is it more suitable than negative transformation?

→ Log transformation increases the brightness of dark regions while preserving image details.

3. Predict the appearance of the enhanced image.

→ The parking lot become brighter

→ Objects such as vehicles, people and road markings become more visible.

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3. Fingerprint Recognition:

→ A fingerprint image has low contrast between ridges and valleys.

1. Which gray-level transformation would you apply?

→ Contrast stretching

2. Explain why.

→ Contrast stretching increases the difference between ridge and valley intensities, making fingerprint patterns sharper and easier to recognize.

3. What would happen if a log transformation were applied instead?

Log transformation mainly enhances dark pixels and compresses brighter pixels.

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